

Christopher Musco

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Education

Massachusetts Institute of Technology

Ph.D. Candidate, Computer Science – Theory of Computation

Advisor: Jonathan Kelner

Masters Thesis: Dimensionality Reduction for Sparse and Structured Matrices (Spring 2015)

Cambridge, MA

2013 – present

Yale University

B.S. Applied Mathematics, B.S. Computer Science

Applied Mathematics Thesis: Fast Discrete Laplace Transforms (with Vladimir Rokhlin)

Computer Science Thesis: Graph Constructions for Machine Learning (with Daniel Spielman)

New Haven, CT

2008 – 2012

Research Areas

theory of algorithms • scalable machine learning • randomized linear algebra • low rank approximation
dimensionality reduction • sketching and streaming • algorithmic graph theory • iterative matrix algorithms

Publications

Authors appear in alphabetical order, in the tradition of mathematics and theoretical computer science.

Conference Papers.....

Input Sparsity Time Low-Rank Approximation via Ridge Leverage Score Sampling. Michael B. Cohen, Cameron Musco, Christopher Musco. *SODA* 2017.

Determining Tournament Payout Structures for Daily Fantasy Sports. Christopher Musco, Maxim Sviridenko, Justin Thaler. *ALLENEX* 2017.

Principal Component Projection Without Principal Component Analysis. Roy Frostig, Cameron Musco, Christopher Musco, Aaron Sidford. *ICML* 2016.

Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition. Cameron Musco, Christopher Musco. *NIPS* 2015. Invited for full oral presentation (1 of 15 out of 403 accepted papers). Also presented at the 2016 Copper Mountain Conference on Iterative Methods

Dimensionality Reduction for K-Means Clustering and Low Rank Approximation. Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *STOC* 2015.

Principled Sampling for Anomaly Detection. Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiropoulos-Douskos, Martin Rinard. *NDSS* 2015.

Uniform Sampling for Matrix Approximation. Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *ITCS* 2015.

Single Pass Spectral Sparsification in Dynamic Streams. Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *FOCS* 2014. Invited to SICOMP Special Issue.

Other.....

Recursive Sampling for the Nyström Method. Cameron Musco, Christopher Musco. *In submission.*

Random Fourier Features for Kernel Ridge Regression. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *In submission.*

Teaching and Mentorship

Advanced Algorithms (MIT 6.854/18.415)

Teaching Assistant

2016

Held weekly office hours, wrote problem sets, and compiled notes and resources for an updated version of this capstone graduate algorithms course. Wrote guide on topics and expectations for student projects. Prepared and delivered a lecture on linear programming relaxations.

Technical Communication Skills for Graduate Students (MIT 6.S977)

Workshop Leader

2016

Lead weekly workshops that supplemented lecture material with hands-on writing and presentation exercises. Met with students one-on-one outside of class to offer individual support and coaching in technical communication.

MIT Graduate Communication Lab

Communication Advisor

2015 – present

Helping to establish an EECS Communication Lab to provide over 700 graduate students with support in academic writing, oral presentation, job applications, and generally, in communicating effectively. Offer feedback, guidance, and coaching to individual students during weekly office hours (75+ individual appointments to date).

CSAIL Algorithms Office Hours

Student Member

2016 – present

Meet with researchers, generally in applied fields, who seek advice on solving or framing algorithmic problems that appear in their research.

Research Science Institute at MIT

Summer Research Advisor

2014

Mentored Christopher Wang, a high school student participating in the prestigious RSI summer program. Chris' project, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

Introduction to Computer Science (Yale CPSC-201)

Teaching Assistant

2012

Covered functional programming, boolean algebra, formal language theory, and basic complexity theory.

Introduction to Programming (Yale CPSC-112)

Teaching Assistant

2011

Led weekly section for 15-25 students covering basic Java programming. Tutored 6 students individually.

Introductory Calculus (Yale MATH-112 & MATH-115)

Grader

2009 – 2010

Scored problem sets and provided explanations and solutions for students, as well as feedback to the professor.

Industry Research

Yahoo Labs

New York, NY

Research Intern, Scalable Machine Learning Group

Summer 2015

Mentors: Justin Thaler, Maxim Srividenko, Edo Liberty

Worked on theoretical matrix sampling problems and on algorithms for managing Yahoo's Fantasy Sports platform.

Talks

Algorithms for Determining Tournament Payout Structures

Meeting on Algorithm Engineering and Experiments (ALENEX)

Jan. 2017

Introduction to Linear Sketching

MIT, Theory Retreat

Sept. 2016

Principal Component Projection without Principal Component Analysis

International Conference on Machine Learning (ICML)	<i>June 2016</i>
Ridge Leverage Score Sampling	
National Institute of Informatics, Shonan Meeting	<i>July 2016</i>
MIT, Algorithms & Complexity Seminar	<i>July 2016</i>
University of Utah, Data Group Meeting	<i>Jan. 2016</i>
Randomized Block Krylov Methods	
Neural Information Processing Systems (NIPS) Poster Session	<i>Dec. 2015</i>
Large Scale Column Subset Selection	
Yahoo Labs, Science Week Poster Session	<i>July 2015</i>
Dimensionality Reduction for K-Means Clustering	
IBM T.J. Watson, Mathematical Sciences Research Seminar	<i>Aug. 2015</i>
Symposium on Theory of Computing (STOC)	<i>June 2015</i>
MIT, Theory Lunch	<i>June 2014</i>
Principled Sampling for Anomaly Detection	
MIT, Defense Advanced Research Projects Agency Site Visit	<i>Mar. 2015</i>
Network and Distributed System Security Symposium (NDSS)	<i>Feb. 2015</i>
Uniform Sampling for Matrix Approximation	
Yahoo Labs, Scalable Machine Learning Research Seminar	<i>June 2015</i>
Single Pass Spectral Sparsification in Dynamic Streams	
MIT, Annual Sublinear Algorithms Day Poster Session	<i>Apr. 2015</i>
Harvard University, Theory Seminar	<i>Nov. 2014</i>
Foundations of Computer Science (FOCS)	<i>Oct. 2014</i>

Service

External Reviewer.....	
SIAM Journal on Matrix Analysis and Applications (SIMAX)	<i>2017</i>
IEEE Transactions on Knowledge and Data Engineering (TKDE)	<i>2017</i>
Conference on Learning Theory (COLT)	<i>2016</i>
Symposium on Discrete Algorithms (SODA)	<i>2016, 2017</i>
IEEE Transactions on Signal Processing (TSP)	<i>2016, 2017</i>
Neural Information Processing Systems (NIPS)	<i>2015, 2016</i>
European Symposia on Algorithms (ESA)	<i>2015</i>
Foundations of Computer Science (FOCS)	<i>2015</i>
International Colloquium on Automata, Languages and Programming (ICALP)	<i>2015, 2017</i>
Network and Distributed System Security Symposium (NDSS)	<i>2015</i>

Other Research Experience

Japan National Institute of Informatics	<i>2016</i>
<i>Invited participant, Shonan meeting on Recent Advances in Randomized Numerical Linear Algebra</i>	

Honors and Awards

National Science Foundation <i>Graduate Research Fellowship, recipient.</i>	2014-2017
Facebook Research <i>Ph.D. Fellowship, finalist (40/800 applicants)</i>	2017
Yale University <i>Cum Laude, Distinction in Both Majors.</i>	2012

Professional Experience

Redfin <i>Software Engineer, Data Team</i> Developed backend Java infrastructure for internet-powered real estate startup. Incorporated large-scale machine learning algorithms into production code. Managed three interns and onboarded four new-hires.	Seattle, WA 2012 – 2014
Elysium Digital <i>Summer Consultant</i> Provided technical expertise in software patent cases at leading litigation consulting company.	Cambridge, MA Summer 2011
Amicus <i>Software Developer</i> Built tools for political campaign management and social fundraising at an early-stage startup that went on to raise nearly 4 million dollars in seed funding.	New Haven, CT 2010 – 2011
Moody's Investors Service <i>Intern, Asset Finance Group</i> Established and monitored bond ratings on securitizations in three asset classes. Wrote code to automate analysis for student loan data and to compile records on aircraft leases.	New York, NY Summer 2009
U.S. Department of Agriculture <i>Intern, Natural Resources Conservation Service</i> Collected and analyzed data for two major state conservation and research projects. Integrated mapping software and database tools to develop a system for compiling and presenting soil data.	East Greenwich, RI Summer 2007